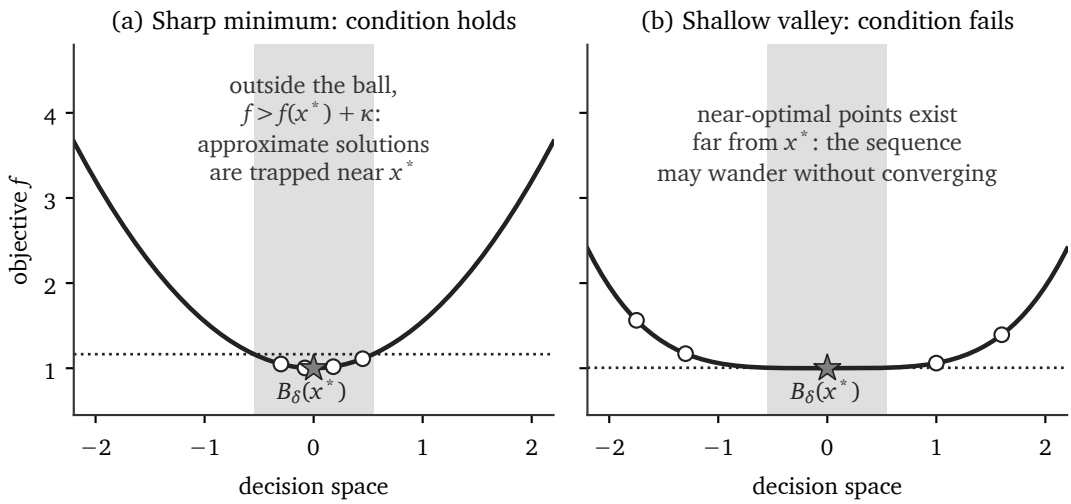
(a) Sharp minimum: condition holds
(b) Shallow valley: condition fails
outside the ball,
$f > f(x^*) + \kappa$:
approximate solutions
are trapped near x^*
near-optimal points exist
far from x^*: the sequence
may wander without converging
objective f
4
3
2
1
$B_\delta(x^*)$
$B_\delta(x^*)$
−2
−1
0
1
2
decision space
decision space